

Omics for Non-Biologists Session 4, Emily's Life on the Line: Choosing the Right Drug Using Multi-Omics

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1 Overview

This report summarizes Emily's multi-omics profile and aligns it with predefined response criteria for three candidate therapies. The analysis spans **genomics**, **transcriptomics**, **proteomics**, **metabolomics**, **ATAC-Seq (chromatin accessibility)**, and **microbiomics** to compute per-drug condition matches and derive a final recommendation.

Eligibility Criteria for Treatments:

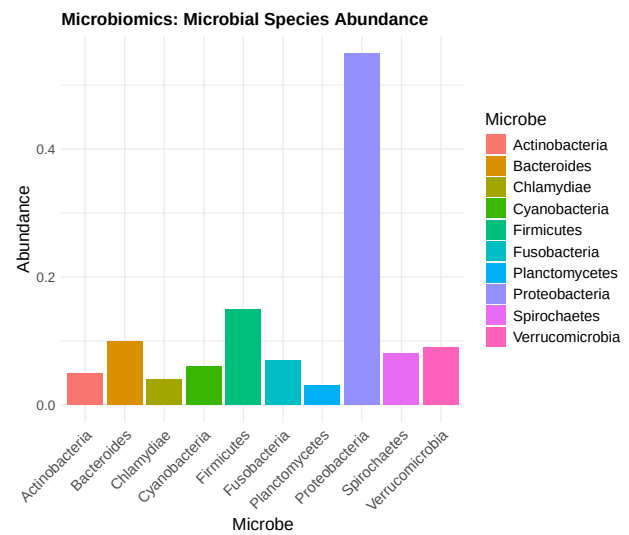
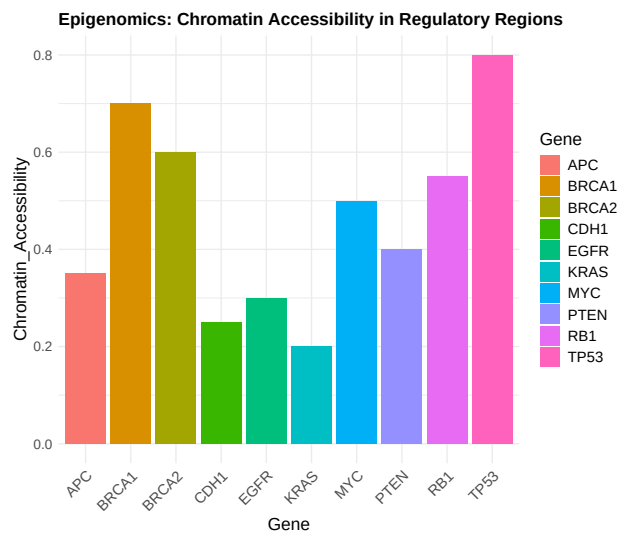
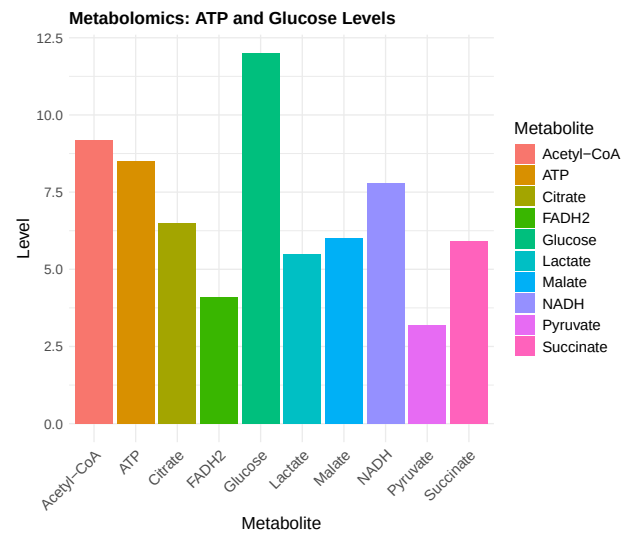
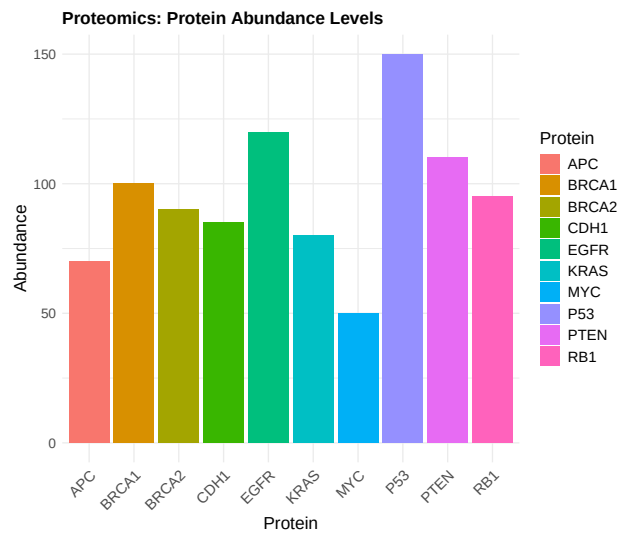
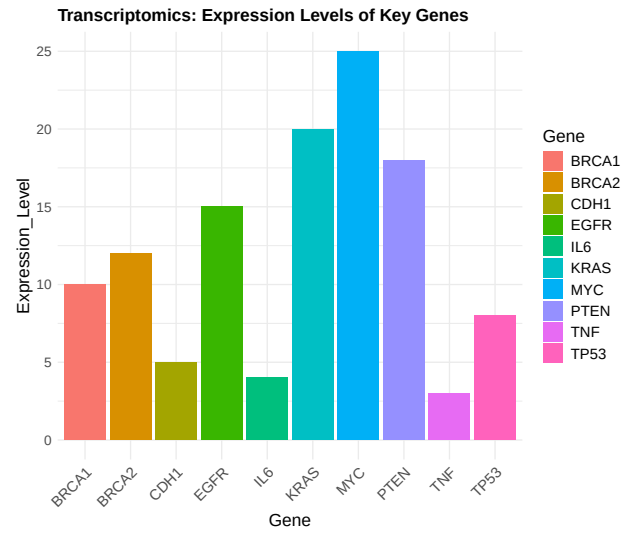
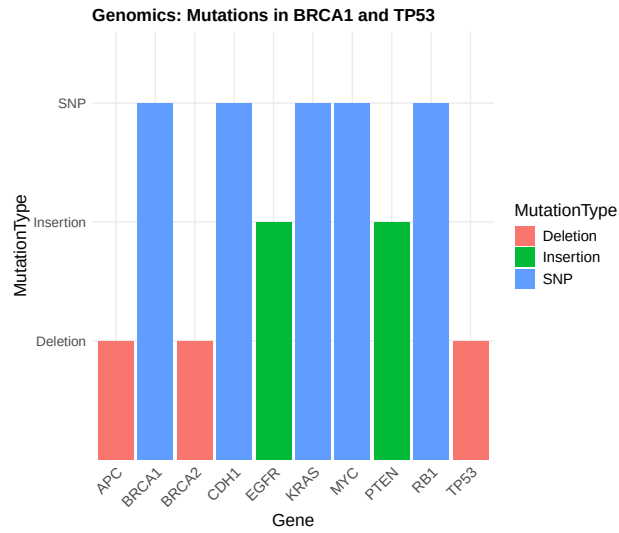
- **NeoThera**: BRCA1 mutation present; TP53 mutation **absent**; MYC > 20; ATP > 8
- **OncoCure**: TP53 mutation present; P53 abundance > 100; KRAS accessibility < 0.3
- **ImmunoPro**: IL-6 < 5; TNF < 5; Glucose > 10; Proteobacteria > 0.5

2 Data Snapshot

##	Dataset	Rows	Cols
##	<chr>	<int>	<int>
## 1	genomics_data	10	3
## 2	transcriptomics_data	10	2
## 3	proteomics_data	10	2
## 4	metabolomics_data	10	2
## 5	epigenomics_data	10	2
## 6	microbiomics_data	10	2

3 Plots

All six omics visualizations are arranged below for side-by-side comparison.



4 Candidate Conditions by Omics

4.1 Genomics

- **BRCA1 mutation present:** Yes
- **TP53 mutation present:** Yes

4.2 Transcriptomics

- **MYC expression:** 25 (threshold > 20)
- **IL-6 expression:** 4 (threshold < 5)
- **TNF expression:** 3 (threshold < 5)

4.3 Proteomics

- **P53 abundance:** 150 (threshold > 100)

4.4 Metabolomics

- **ATP level:** 8.5 (threshold > 8)
- **Glucose level:** 12 (threshold > 10)

4.5 ATAC-Seq

- **KRAS chromatin accessibility:** 0.2 (threshold < 0.3)

4.6 Microbiomics

- **Proteobacteria abundance:** 0.55 (threshold > 0.5)

5 Drug Scoring

##	Drug	Conditions	Matched	Total	Percent	MeetsAll
##	<chr>	<chr>	<int>	<int>	<dbl>	<lgl>
## 1	ImmunoPro	IL6_low: Yes TNF_low: Yes Glucos~	4	4	100	TRUE
## 2	OncoCure	TP53_mut_present: Yes P53_abundanc~	3	3	100	TRUE
## 3	NeoThera	BRCA1_mut_present: Yes TP53_mut_ab~	3	4	75	FALSE

6 Recommendation

Recommended drug: ImmunoPro

7 Multi-omics Framework for Therapy Selection

Rule-Based Decision Logic

1. **Strict match preference:** If any drug satisfies *all* of its molecular conditions, it is recommended. This enforces a bias toward therapies whose efficacy is supported by the strongest concordant evidence across multiple omics layers.
2. **Relative strength:** If no therapy fully satisfies its criteria, the drug(s) with the highest percentage of matched conditions are selected. This ensures that even partial matches can still inform a clinically relevant choice.
3. **Handling ties:** If multiple therapies remain indistinguishable after steps 1 and 2, the one with the greater total number of conditions is selected. However, if two treatments achieve a perfect match, then a tie is declared, signaling the need for additional data, clinical judgment, or external evidence to adjudicate the decision.

Clinical Rationale Summary

Emily’s multi-omics profile was systematically aligned with predefined molecular criteria for three candidate therapies. The decision framework is intentionally transparent and rule-based.

- This approach demonstrates how heterogeneous datasets can be distilled into actionable decision rules, allowing us to bridge molecular biology and clinical choice.
- While the rules applied here are simplified, the workflow illustrates the broader goal of *precision medicine*: tailoring custom interventions to the patient’s unique molecular landscape.
- In practice, such frameworks would be complemented with statistical validation, external cohorts, and clinical endpoints, but even in simplified form they highlight the power of **multi-omics integration** for evidence-based therapy selection.

Results Interpretation

While the computational scoring framework provided a clear ranking of candidate therapies, the interpretation of these results requires caution.

- **NeoThera (Drug A)** is unsuitable because it explicitly requires the *absence* of a TP53 mutation, yet Emily’s tumor carries a TP53 deletion. This automatically disqualifies it from consideration.
- **OncoCure (Drug B)** and **ImmunoPro (Drug C)** both meet their molecular criteria within Emily’s profile. From a rule-based perspective, Drug C emerged as the statistically stronger match. However, Drug B aligns with the TP53 mutation, elevated P53 abundance, and low KRAS accessibility, features that point directly to the tumor’s genetic drivers.
- **Drug C**, by contrast, satisfies immune-metabolic criteria (low IL-6 and TNF, high glucose, dominance of Proteobacteria). These signals suggest potential value in stabilizing immune or metabolic balance, but they do not address the underlying oncogenic mechanism. Thus, it may instead play a supportive or adjunctive role rather than serving as the primary therapeutic anchor.